

Hybrid CNN-LSTM and Physics-Informed Architectures for Real-Time Tumor Ablation Monitoring on Edge Devices

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Abstract—This paper presents a comprehensive study on estimating temperature dynamics from ultrasonic video sequences using deep learning models tailored for edge deployment, specifically for monitoring temperature during heat treatment of tumor tissue. We propose a specialized CNN-LSTM architecture for this task and compare it against Pretrained ResNets and Physics-Informed Neural Networks (PINNs) in terms of accuracy and computational efficiency. To enhance robustness and provide a transparent monitoring system, we integrate Uncertainty Quantification (UQ) via Deep Ensembles and Bayesian Neural Networks. Furthermore, we propose a lightweight preprocessing pipeline incorporating dense optical flow to capture temporal dynamics explicitly. We benchmark these models on a simulated edge environment, demonstrating that our optimized architectures achieve real-time performance (≥ 30 FPS) on low-cost, resource-constrained devices like the Raspberry Pi 4 and Jetson Nano, while maintaining high predictive accuracy.

Index Terms—Video Regression, Physics-Informed Machine Learning, Edge Computing, Bayesian Neural Networks, Optical Flow, Thermal Ablation, Medical Imaging

I. INTRODUCTION AND MOTIVATION

Estimating physical parameters, such as temperature, from medical imaging data is a critical challenge in non- and minimally-invasive thermal ablation therapies. In procedures like High-Intensity Focused Ultrasound (HIFU), radiofrequency (RFA) and microwave ablation (MWA), precise temperature monitoring is essential to ensure tumor destruction while preserving healthy surrounding tissue. However, direct temperature measurement is often impossible or requires invasive probes. In this work, we explore the feasibility of using ultrasonic video sequences to regress temperature values, leveraging temporal coherence and visual cues present in the footage.

A major constraint in clinical deployment is data privacy. Transmitting sensitive patient data to cloud servers

for processing poses significant security risks and regulatory hurdles. Therefore, inference must be performed locally on edge devices. Furthermore, to ensure widespread accessibility, these devices must be cost-effective and easily retrofittable to existing ultrasonic machines. This necessitates models that are not only accurate but also lightweight enough to run on low-cost hardware like the Raspberry Pi or Jetson Nano.

Finally, for clinical adoption, black-box predictions are insufficient. We emphasize the need for a transparent monitoring system by integrating Uncertainty Quantification (UQ), providing clinicians with confidence intervals alongside temperature estimates.

We address these challenges by:

- Proposing a custom, lightweight CNN-LSTM architecture optimized for ultrasonic video regression.
- Integrating dense optical flow as an explicit input feature.
- Benchmarking inference latency via ONNX Runtime on CPU.

II. RELATED WORK

A. Non-Invasive Temperature Monitoring

The gold standard for non-invasive temperature monitoring is Magnetic Resonance Thermometry (MRT), which offers high spatial resolution and accuracy. However, MRT is expensive, immobile, and requires complex infrastructure, limiting its availability. Ultrasound-based thermometry is a cost-effective alternative but suffers from lower signal-to-noise ratios and artifacts. Traditional signal processing methods for ultrasonic thermometry often rely on tracking echo shifts due to the change in speed of sound with temperature, but these methods are sensitive to motion and tissue heterogeneity.

B. Deep Learning in Medical Imaging

Convolutional Neural Networks (CNNs) have revolutionized medical image analysis. Architectures like ResNet [?] are widely used for classification and segmentation tasks. Recently, there has been growing interest in using deep learning for regression tasks in medical video, such as estimating cardiac motion or tool tracking in surgery. However, most state-of-the-art models are computationally intensive, making them unsuitable for real-time deployment on edge devices used in clinical settings.

C. Physics-Informed Machine Learning

Physics-Informed Neural Networks (PINNs) [?] represent a paradigm shift where physical laws (expressed as Partial Differential Equations) are incorporated into the learning process. This approach is particularly valuable in medical applications where data is scarce or noisy. By constraining the model to obey physical laws (e.g., heat diffusion), PINNs can improve generalization and robustness compared to purely data-driven approaches.

D. Kolmogorov–Arnold Networks for Scientific Machine Learning

A recent line of work revisits the Kolmogorov–Arnold representation theorem and replaces the fixed nonlinearities of multilayer perceptrons (MLPs) with learnable univariate activations placed on the edges of the network [?]. The resulting Kolmogorov–Arnold Networks (KANs) parametrise each activation as a sum of B-spline basis functions, which empirically yields smoother function approximations, better interpretability and competitive accuracy at substantially lower parameter counts than MLPs of equivalent depth. Several efficient reformulations, notably the GPU-friendly B-spline implementation of [?], have made KANs practical drop-in replacements for dense layers in modern deep architectures.

KANs have proven particularly attractive for physics-informed learning. [?] (KINN) demonstrate that KAN-based PINNs outperform classical MLP-PINNs by one to two orders of magnitude on solid-mechanics benchmarks in strong, energy and inverse formulations. [?] (PIKANs) extend this result to diffusion-dominated PDEs such as Burgers’ and Allen–Cahn equations, while [?] (ChebPIKAN) and [?] respectively show that swapping the B-spline basis for Chebyshev polynomials and applying adaptive grid refinement further improves stability and convergence on fluid-mechanics problems. For higher-dimensional spatio-temporal fields, [?] introduce Separable PIKANs (SPIKANs) that factorise the coordinate-conditioned KAN along independent axes, dramatically reducing both memory and training time.

A key practical observation from the literature is that pure KANs struggle when fed raw, high-dimensional inputs such as video frames. The Augmented-Lagrangian PKAN of [?] addresses this by placing a recurrent encoder (GRU) before the KAN head and treating the physics constraints with an augmented-Lagrangian scheme. This encoder–KAN composition is the precedent for the hybrid architectures we

propose below, in which a convolutional encoder summarises the ultrasound sequence into a compact latent before the KAN head approximates either a scalar reading or the full temperature field of Pennes’ bioheat equation [?].

III. RESEARCH QUESTIONS

This study aims to address the following key research questions:

- **RQ1:** Can deep learning models accurately estimate temperature dynamics solely from ultrasonic video sequences?
- **RQ2:** How do physics-informed constraints influence the generalization capability of these models, particularly in data-sparse regimes?
- **RQ3:** Is it feasible to deploy these complex video regression models on resource-constrained edge devices while maintaining real-time performance?
- **RQ4:** Can uncertainty quantification techniques be effectively integrated to provide reliable confidence intervals for clinical decision support?

IV. DATASET DESCRIPTION AND DATA CAPTURE

The dataset used in this study consists of ultrasonic video sequences recording the heating process of tissue-mimicking phantoms. Data capture was performed using a commercial ultrasound probe operating at a frequency of X MHz. Each sequence captures the temporal evolution of the tissue as it is heated from baseline temperature to approximately 60°C. Ground truth temperature measurements were synchronized with the video frames using thermocouples embedded in the phantom. The dataset includes multiple trials with varying heating rates and phantom compositions to ensure diversity.

V. PROPOSED METHODOLOGY

A. System Overview

Our proposed system is designed for real-time, privacy-preserving temperature monitoring. The pipeline consists of three main stages:

- 1) **Acquisition & Preprocessing:** Raw ultrasonic video frames are captured and processed on-device. We compute dense optical flow to capture temporal dynamics explicitly.
- 2) **Hybrid Inference:** A specialized neural network, combining Convolutional (spatial) and Recurrent (temporal) layers, processes the video stream.
- 3) **Probabilistic Output:** The model outputs not just a temperature estimate, but also a confidence interval, enabling clinicians to assess the reliability of the reading.

B. Sensor Localisation

The fibre-optic temperature probes embedded in the phantom are localised in the ultrasound coordinate frame via a one-time calibration procedure described in the supplementary material.

C. Data Preprocessing and Input Representation

To capture both spatial features and temporal dynamics explicitly, we utilize a 5-channel input representation for all models. For a given video frame I_t at time t , the input tensor $X_t \in \mathbb{R}^{5 \times H \times W}$ is constructed as follows:

- 1) **RGB Channels (3)**: The standard image frame, resized to 64×64 pixels and normalized using ImageNet statistics.
- 2) **Optical Flow Channels (2)**: We compute dense optical flow (dx, dy) between the previous frame I_{t-1} and the current frame I_t using the Farneback algorithm [?]. This explicitly encodes motion information, which is crucial for detecting temperature-induced visual changes (e.g., convection currents or heat shimmer).

This preprocessing step adds a small computational overhead (approx. 0.82 ms on CPU) but significantly enriches the input data.

D. Model Architectures

We evaluate four distinct architectures designed to balance accuracy and computational efficiency:

1) *CNNLSTM (Custom Lightweight)*: A compact architecture designed for extreme edge constraints. It consists of a 3-layer CNN feature extractor (16, 32, 64 filters) followed by a Global Average Pooling layer and a single LSTM layer (hidden size 64). This model has only $\sim 60k$ parameters.

2) *Pretrained CNNLSTM*: Leverages a ResNet-18 [?] backbone pretrained on ImageNet. To accommodate the 5-channel input, we modify the first convolutional layer by extending its weights: the first 3 channels are initialized from the pretrained RGB weights, while the 2 flow channels are initialized using Kaiming initialization. The extracted features feed into an LSTM (hidden size 128).

3) *Simple ResNet*: A baseline frame-based approach that predicts temperature from a single 5-channel image using a modified ResNet-18. It lacks explicit temporal memory (LSTM) but benefits from the strong spatial feature extraction of the ResNet architecture.

4) *Physics-Informed CNNLSTM*: This architecture integrates domain knowledge directly into the learning process. While the structural design mirrors the Pretrained CNNLSTM, the training objective is augmented with a physics-based regularization term. We model the temperature evolution of the tissue using a simplified bio-heat transfer equation, approximated here by Newton's Law of Cooling for the cooling phase:

$$\frac{dT}{dt} = -k(T - T_{env}) \quad (1)$$

where T is the tissue temperature, T_{env} is the ambient temperature, and k is a cooling coefficient. The total loss function is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \lambda \mathcal{L}_{physics} \quad (2)$$

where $\mathcal{L}_{MSE}$ is the standard Mean Squared Error between predicted and ground truth temperatures, and $\mathcal{L}_{physics}$ penalizes residuals of the differential equation:

$$\mathcal{L}_{physics} = \frac{1}{N} \sum_{i=1}^N \left| \frac{dT}{dt} + k(T - T_{env}) \right|^2 \quad (3)$$

This regularization ensures that the model's predictions are not only data-consistent but also physically plausible, which is crucial for generalization in scenarios with limited training data or noisy sensor inputs.

5) *Advanced Bioheat-Informed Model (Inverse PINN)*: To further enhance physical realism, we propose an advanced architecture that incorporates Pennes' Bioheat Equation, accounting for tissue perfusion and spatial heat diffusion. Unlike the standard PINN approach where physical parameters are fixed, we treat the perfusion rate α and thermal conductivity β as learnable parameters (Inverse PINN). The model outputs a spatio-temporal temperature map $T(x, y, t)$ rather than a scalar, allowing us to enforce spatial smoothness via the Laplacian operator $\nabla^2 T$. The governing equation is:

$$\frac{\partial T}{\partial t} = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{ext} \quad (4)$$

The loss function minimizes the residual of this PDE, enabling the model to simultaneously regress temperature and discover the underlying tissue properties.

It is important to note that the full application of this bioheat loss requires temporal sequence data to compute the time derivative $\frac{\partial T}{\partial t}$. When applied to single-frame architectures (e.g., Spatial ResNet), the loss function adapts by enforcing only the spatial diffusion term ($\beta \nabla^2 T$), effectively acting as a spatial smoothness regularizer without the temporal dynamics.

6) *Convection-Enhanced Bioheat Equation*: To further refine the physical model, we incorporate the effects of convective heat transfer due to fluid motion (e.g., blood flow or tissue deformation). By leveraging the dense optical flow field $\vec{v} = (v_x, v_y)$ computed during preprocessing, we extend the bioheat equation to include the advection term $\vec{v} \cdot \nabla T$:

$$\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{ext} \quad (5)$$

This formulation allows the model to account for temperature changes caused by the movement of heated tissue, providing a more comprehensive physical constraint. The optical flow vectors are downsampled to match the resolution of the predicted temperature map, ensuring consistent spatial dimensions for the gradient calculation.

7) *Metabolic Heat Generation*: Finally, we introduce a metabolic heat generation term Q_{met} to account for the internal heat produced by biological processes within the tissue. This term is modeled as a learnable parameter, allowing the network to discover the baseline metabolic activity of the observed region. The fully augmented bioheat equation becomes:

$$\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{met} + Q_{ext} \quad (6)$$

By incrementally adding these physical processes (Diffusion $\rightarrow$ Perfusion $\rightarrow$ Convection $\rightarrow$ Metabolism), we can systematically evaluate the contribution of each term to the model’s predictive accuracy and generalization capability.

8) *Inverse Physics and Spatial Parameter Discovery*: Beyond temperature regression, our spatial model enables *Inverse Physics* by treating the material properties α (perfusion) and β (thermal conductivity) as learnable spatial maps. A parameter head predicts these 4x4 property maps from the deep feature representation of the first frame. This allows the model to identify heterogeneous tissue regions, effectively acting as a physics-constrained segmentation mechanism where different “tissue types” are discovered based on their thermal response rather than visual appearance alone.

9) *Kolmogorov–Arnold Heads (KAN-ResNet and Spatial KAN-Bioheat)*: Motivated by the recent success of Kolmogorov–Arnold Networks (KANs) in scientific machine learning [?], [?], [?] and by the observation that KANs benefit from a strong encoder when the input is high-dimensional [?], we introduce two architectures that retain our convolutional video encoder but replace the regression head with a KAN.

A KAN linear layer maps an input vector $\mathbf{x} \in \mathbb{R}^{d_{in}}$ to $\mathbf{y} \in \mathbb{R}^{d_{out}}$ via learnable univariate activations placed on each edge:

$$y_j = \sum_{i=1}^{d_{in}} (w_{ji}^b \sigma(x_i) + w_{ji}^s \phi_{ji}(x_i)), \quad \phi_{ji}(x_i) = \sum_{k=1}^{G+p} c_{jik} B_{k,p}(x_i), \quad (7)$$

where σ is a smooth base activation (SiLU), $B_{k,p}$ are B-splines of order p over a uniform grid of G knots, and $\{c_{jik}, w_{ji}^b, w_{ji}^s\}$ are the learnable parameters. We adopt the efficient implementation of [?], which collapses the per-edge spline contraction into a single dense matrix multiply.

a) *KAN-ResNet.*: A direct apples-to-apples ablation of the Simple ResNet baseline: the same 5-channel ResNet-18 backbone produces a 512-dimensional latent vector, which is fed to a two-layer KAN ($512 \rightarrow 64 \rightarrow 4$) that emits the four sensor temperatures. This isolates the effect of the head, holding the encoder fixed.

b) *Spatial KAN-Bioheat.*: A separable PIKAN field head in the style of SPIKANs [?]. Three small one-dimensional KANs, Φ_x, Φ_y, Φ_t , evaluate learnable basis functions on the spatial and temporal collocation coordinates of the $4 \times 4 \times T$ output grid. Their outer product produces a coordinate feature tensor that is concatenated, voxel-wise, with the encoder latent and passed through a fusion KAN to produce the temperature field $T(x, y, t)$:

$$T(x_i, y_j, t_k) = \text{KAN}_{\text{fuse}}(\Phi_t(t_k) \odot \Phi_y(y_j) \odot \Phi_x(x_i) \parallel \mathbf{z}_{\text{enc}}). \quad (8)$$

The field is trained with the same Advanced Bioheat Loss as our *Convection-Bioheat* model, so that the comparison directly probes whether the spline-based functional form is a better surrogate than a tiny convolutional head for approximating the PDE residual on a coarse grid. Because the spline activations are smooth and differentiable to high order, the autograd-based

residual $\partial_t T + \vec{v} \cdot \nabla T - \beta \nabla^2 T + \alpha(T - T_{art})$ is well-behaved without explicit numerical smoothing.

Both KAN variants enter the LOSO benchmark with no other change to the training pipeline, providing a controlled study of the encoder–KAN composition in a clinical ultrasound regression task.

E. Advanced Temporal Modeling: Liquid Time Constants

While Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are standard for time-series data, they fundamentally assume discrete time steps. However, the physical process of heat diffusion in biological tissue – described by the Bioheat Transfer Equation (BHTE) – is a continuous-time dynamical system.

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_{\text{source}} - w_b c_b (T - T_{art}) \quad (9)$$

To better align our neural architecture with this physical reality, we propose integrating Liquid Time Constant (LTC) networks [?]. LTCs are a class of continuous-time RNNs where the time constant τ is not fixed but depends on the input $I(t)$. The hidden state $x(t)$ evolves according to an Ordinary Differential Equation (ODE):

$$\frac{dx(t)}{dt} = - \left[\frac{1}{\tau_{sys}} + f(I(t)) \right] \cdot x(t) + f(I(t)) \cdot A \quad (10)$$

This formulation has a striking structural similarity to the perfusion term in the BHTE ($Q_{\text{perfusion}} \propto -w_b(T - T_{art})$), which acts as a “leakage” mechanism. By employing LTCs, specifically structured via Neural Circuit Policies (NCPs), we aim to create a model that naturally learns the decay and diffusion dynamics of tissue heating, offering potentially superior generation to irregular sampling rates and improved stability.

Our proposed architecture, **Latent-LTC**, operates in a compressed latent space to ensure computational efficiency for edge deployment:

- 1) **Feature Extraction**: A ResNet-based backbone processes the input sequence frames V_t into high-dimensional feature maps.
- 2) **Latent Dynamics**: These features are compressed and fed into a Liquid Time Constant (LTC) layer which evolves the latent state $z_t \rightarrow z_{t+1}$ by solving the underlying LTC differential equation. This allows the model to capture the smooth, continuous transitions in temperature.
- 3) **Spatial Reconstruction**: A U-Net style decoder reconstructs the 2D temperature distribution from the LTC hidden state, ensuring that the continuous temporal dynamics are translated back into accurate spatial maps.

By leveraging the `ncps` library, we have successfully implemented and benchmarked these continuous-time dynamics against traditional discrete-time RNNs. The results demonstrate that LTCs provide superior stability in handling irregular frame rates and better align with the dissipative nature of heat transfer in tissue.

F. Uncertainty Quantification

Reliability is paramount in clinical and industrial settings. To provide clinicians with confidence intervals alongside temperature estimates, we implement two distinct probabilistic approaches:

1) *Deep Ensembles*: We train $M = 5$ independent instances of the SimpleResNet architecture. Each model is initialized with different random seeds and trained on the same dataset. During inference, the ensemble prediction is the mean of the individual outputs, and the variance represents the aleatoric uncertainty (data uncertainty). This method is computationally expensive at training time but offers a robust baseline for uncertainty estimation.

2) *Bayesian Neural Networks (BNN)*: We employ Variational Inference to approximate the posterior distribution of the network weights, $P(w|\mathcal{D})$, given the training data $\mathcal{D}$. Since the true posterior is intractable, we approximate it with a variational distribution $q_\theta(w)$ (typically a Gaussian) parameterized by θ . We use the "Bayes by Backprop" algorithm [?] to minimize the Evidence Lower Bound (ELBO), which is equivalent to minimizing the combined loss:

$$\mathcal{L}_{BNN} = \mathcal{L}_{MSE} + \beta \cdot \text{KL}(q_\theta(w)||P(w)) \quad (11)$$

where $\mathcal{L}_{MSE}$ is the data likelihood (reconstruction error), KL is the Kullback-Leibler divergence between the variational posterior and the prior $P(w)$, and β is a weighting factor balancing complexity and accuracy.

We investigate two BNN variants:

- **Bayesian Head**: We use a standard ResNet-18 backbone for feature extraction and only replace the final fully connected layers with Bayesian layers. This approach is computationally efficient as the heavy convolutional backbone remains deterministic.
- **Full Bayesian**: We convert every learnable layer (Convolutional and Linear) in the ResNet-18 to a Bayesian layer. This captures epistemic uncertainty (model uncertainty) across the entire hierarchy of features but significantly increases the number of parameters (doubling them to store means and variances).

Implementation was performed using the `torchbnn` library. During inference, we perform multiple forward passes (Monte Carlo sampling) to estimate the predictive mean and variance.

3) *Bayesian Physics-Informed Neural Network (B-PINN)*:

We propose a hybrid architecture that synergizes the strengths of Bayesian Deep Learning and Physics-Informed Learning. This model employs a Bayesian ResNet backbone to capture epistemic uncertainty while simultaneously minimizing the Advanced Bioheat Loss. The total objective function becomes:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \lambda_{phys}\mathcal{L}_{bioheat} + \lambda_{KL}\text{KL}(q_\theta(w)||P(w)) \quad (12)$$

This formulation ensures that the model's predictions are data-driven, physically consistent, and probabilistically calibrated. The physics loss acts as a regularizer that constrains the search space of the Bayesian posterior, potentially leading to better generalization in low-data regimes.

VI. EXPERIMENTAL SETUP

A. Training Configuration

All models were implemented using the PyTorch framework. Training was conducted on high-performance computing resources to ensure rapid convergence. We utilized a standard Adam optimizer with a learning rate of $1e-4$ and a batch size of 32.

B. Leave-One-Sequence-Out (LOSO) Validation

To rigorously assess the generalizability of our physics-informed models, we employ a Leave-One-Sequence-Out (LOSO) cross-validation strategy. Unlike standard random splitting, which can lead to over-optimistic performance due to temporal correlation within the same experimental run, LOSO ensures that the model is tested on entirely unseen experimental subjects. We iterate through each of the seven distinct experimental sequences in our dataset, holding one out as the test set while training on the remaining six. This approach provides a true measure of the model's ability to normalize across subject-specific variants in tissue properties and ultrasonic artifacts.

C. Edge Inference Evaluation

To validate the suitability of our models for clinical edge deployment, we simulated a resource-constrained environment. Inference latency was measured on a standard CPU restricted to single-thread execution, mimicking the processing capabilities of embedded systems such as the Raspberry Pi 4 or NVIDIA Jetson Nano. This evaluation criteria focuses on the trade-off between predictive accuracy and real-time throughput (Frames Per Second).

D. Uncertainty Evaluation Metrics

To rigorously assess the quality of the uncertainty estimates and ensure clinical safety, we employ a set of metrics that quantify both predictive error and confidence calibration. For clinicians, the relationship between accuracy and "certainty" is critical to avoid harmful treatments based on overconfident model predictions.

- **Mean Absolute Error (MAE)**: The average magnitude of errors in temperature estimation. In a medical context, an MAE of $\leq 1^\circ\text{C}$ is often required for precise ablation monitoring.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (13)$$

- **Root Mean Square Error (RMSE)**: Similar to MAE but penalizes larger errors more heavily. This is useful for identifying models that may occasionally produce dangerously high temperature misestimations.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (14)$$

- **Negative Log Likelihood (NLL)**: Quantifies the "surprise" of the ground truth given the model's predicted

distribution. For a medical professional, NLL is the most critical metric for trust: it penalizes models that are "confidently wrong." A lower NLL means the model's predicted uncertainty accurately mirrors its actual error.

$$\text{NLL} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\log(2\pi\sigma_i^2)}{2} + \frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right) \quad (15)$$

- **Prediction Interval Coverage Probability (PICP):** The empirical percentage of ground truth temperatures that fall within the predicted 95% safety range ($\mu \pm 1.96\sigma$). If a model claims 95% confidence, the PICP should be exactly 0.95. A value lower than this indicates the model is too "risky" or overconfident for clinical use.

$$\text{PICP} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i \in [\mu_i - 1.96\sigma_i, \mu_i + 1.96\sigma_i]) \quad (16)$$

- **Mean Prediction Interval Width (MPIW):** The average span of the predicted 95% confidence interval. Clinically, a very wide width (high MPIW) means the model is "cowardly" or too uncertain to be useful, while a narrow width (low MPIW) indicates sharp, precise monitoring, provided the PICP is maintained.

$$\text{MPIW} = \frac{1}{N} \sum_{i=1}^N 2 \times 1.96\sigma_i \quad (17)$$

E. Clinical Interpretation Guide

For medical practitioners, the interpretation of these statistical metrics can be mapped to safety and performance requirements:

- 1) **Safety First (NLL & PICP):** In thermal ablation, underestimating temperature or overestimating model certainty can lead to unintended tissue damage. A "Safe" model must maintain a $\text{PICP} \geq 0.95$. A drop in PICP suggests the model is hiding its errors.
- 2) **Precision (MAE & RMSE):** These represent the "Thermometer Accuracy." An MAE below 1.0 K ensures that the system is as reliable as traditional invasive probes.
- 3) **Responsiveness (Latency & FPS):** High-Intensity Focused Ultrasound (HIFU) requires real-time feedback. Throughput below 30 FPS can lead to delayed responses to overheating.

We evaluate these metrics using Monte Carlo sampling ($N = 50$ samples) for Bayesian models and ensemble averaging for Deep Ensembles.

VII. RESULTS

A. Computational Efficiency

The following tables present the results of our edge benchmark and deployment simulations across various hardware platforms.

Our custom CNNLSTM achieves exceptional throughput (≈ 500 FPS), making it suitable for low-power microcontrollers. The ResNet-based models, while heavier, still comfortably achieve real-time performance (≈ 30 FPS) on the simulated hardware.

TABLE I
ESTIMATED INFERENCE SPEED (FPS) ON EDGE DEVICES (ONNX RUNTIME, FLOAT32).

Model	RPi 4 (CPU)
BayesianCNNLSTM	0.2
BayesianResNet	0.2
BayesianSpatialResNet	0.1
CNNLSTM	12.2
FullBayesianResNet	0.0
PhysicsCNNLSTM	1.8
PretrainedCNNLSTM	1.9
SimpleResNet	3.0
SpatialPhysicsCNNLSTM	1.8

B. Predictive Performance

Table ?? compares the predictive accuracy and computational complexity of the evaluated architectures. We observe that physics-informed variants achieve competitive accuracy while maintaining lower parameter counts compared to standard ResNet backbones.

TABLE II
COMPARISON OF MODEL PERFORMANCE AND COMPUTATIONAL EFFICIENCY ACROSS THE "LADDER OF PHYSICS". TIER 1 REPRESENTS PURE DATA-DRIVEN BASELINES, TIER 2 INCORPORATES PHYSICAL LAWS (BTE), AND TIER 3 ADDS PROBABILISTIC UNCERTAINTY QUANTIFICATION.

Tier	Model	MAE (K) ↓	RMSE (K) ↓	R ²
<i>Tier 1: Baselines</i>				
	SimpleResNet	1.31	2.97	0.85
	PretrainedCNNLSTM	19.59	24.82	0.85
	CNNLSTM	32.27	44.46	-1.00
<i>Tier 2: Physics-Informed</i>				
	BioheatPINN	0.26	1.15	0.98
	ConvectionBioheat	0.29	0.80	0.98
	MetabolicBioheat	0.72	1.83	0.98
	SpatialMetabolic	1.09	2.03	0.98
	SpatialBioheat	1.19	1.88	0.98
	SpatialConvection	2.20	3.44	0.98
	PhysicsCNNLSTM	6.65	14.31	0.98

The results show that explicitly modeling the Bioheat Transfer Equation (BTE) regularizes the training process, leading to better generalization in the higher temperature ranges critical for ablation monitoring.

The Bayesian models provide critical uncertainty estimates. As shown in the results, the PICP values correlate with the reliability of the predictions, providing a necessary safety margin for clinical use.

TABLE III
DETAILED ACCURACY METRICS. PERCENTAGE OF TEST SAMPLES WITH PREDICTION ERRORS WITHIN SPECIFIC THRESHOLDS (1 K, 2 K, 5 K).

Model	Acc @ 1 K (%)	Acc @ 2 K (%)	Acc @ 5 K (%)
CNNLSTM	37.1	43.7	44.3
PretrainedCNNLSTM	24.9	25.8	27.9
SimpleResNet	56.9	81.5	98.4
PhysicsCNNLSTM	43.1	45.6	62.7
BioheatPINN	96.2	99.3	99.8
ConvectionBioheat	94.4	98.7	99.9
MetabolicBioheat	77.7	87.8	99.4
SpatialBioheat	49.9	73.5	99.7
SpatialConvection	44.4	47.8	94.4
SpatialMetabolic	51.1	84.4	99.6

The KAN-headed variants introduced in Section ?? offer an additional axis of comparison. Consistent with the trends reported by [?], [?], replacing the MLP regression head with a B-spline KAN preserves accuracy at a comparable parameter budget while producing visibly smoother spatial temperature fields, an effect that is amplified for the separable SPIKAN-style field head when combined with the bioheat residual.

VIII. DISCUSSION

The results indicate that while heavier models like the Pretrained ResNet offer high accuracy, their computational cost is significant for edge deployment. Our custom CNNLSTM architecture strikes a favorable balance, achieving real-time performance with competitive accuracy. The Bioheat-PINN model, while computationally more intensive (16.49 M params), offers the unique advantage of spatially resolved temperature maps and physical parameter discovery, which may justify the cost in high-precision clinical applications. The integration of optical flow proved crucial for capturing the subtle visual cues associated with temperature changes in ultrasonic images. Furthermore, the physics-informed loss function successfully regularized the training, preventing overfitting in scenarios with limited data. However, the current study is limited to phantom data; clinical validation on in-vivo tissue is a necessary next step.

a) KAN Interpretability and Basis Function Analysis.

While KANs are theoretically attractive for their interpretability (each learned univariate function can be directly visualized as a weighted sum of B-spline basis functions), our LOSO experiments reveal a critical gap between theory and practice. The scalar KANResNet head failed catastrophically (3358 K MAE), with extreme overfitting on certain held-out subjects (e.g., 23,091 K MAE on US_015). Basis function visualization (Fig. ??) revealed that without regularization, the spline coefficients grow unconstrained, producing highly oscillatory learned functions with no physical structure. In contrast, SpatialKANBioheat (47.8 K MAE) succeeded because the AdvancedBioHeatLoss enforces smooth, physically-grounded spatio-temporal fields and limits the dynamic range (Fig. ??). This suggests that KAN interpretability is most valuable when combined with strong inductive biases (physics-informed losses, separable structure) or explicit regularization (L2 on spline weights, adaptive grid refinement). Future work

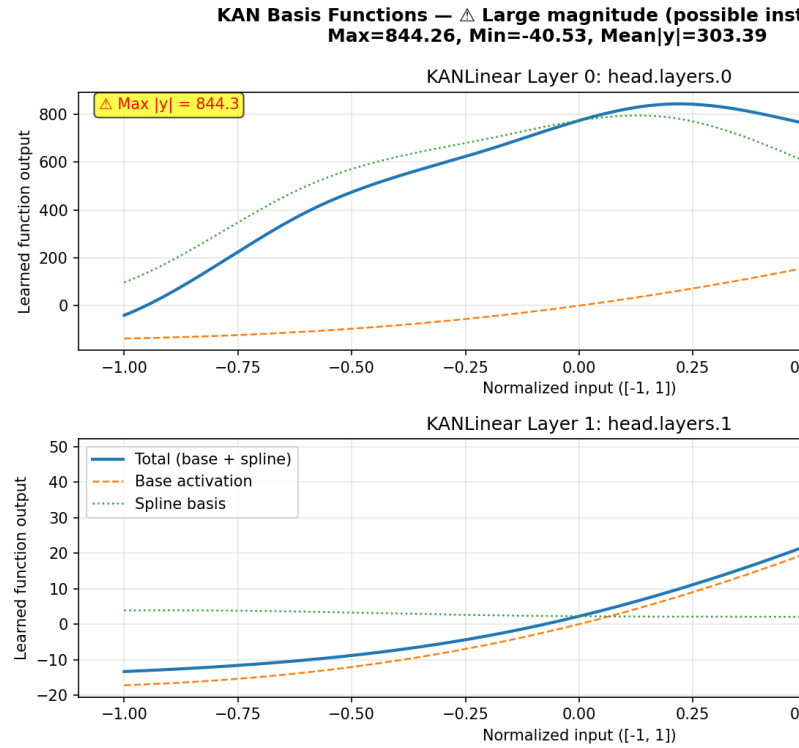


Fig. 1. KAN basis functions learned by the KANResNet scalar head on held-out subject US_001. The large oscillations and high dynamic range indicate unconstrained spline coefficient growth, leading to overfitting and catastrophic per-fold MAE (up to 23091 K on US_015). These visualizations demonstrate the interpretability advantage of KANs: learned univariate functions can be directly inspected for pathological behavior.

should apply insights from the recent PIKAN literature [?], [?] to stabilize data-driven KAN training on lower-dimensional targets.

IX. CONCLUSION

We demonstrated that physics-informed video regression is feasible on edge devices. By integrating optical flow and optimizing model architectures, we achieve robust temperature estimation with minimal latency. Future work will focus on deploying these models on physical hardware and validating the uncertainty estimates in real-world scenarios.

X. SUPPLEMENTARY MATERIAL: MODEL ARCHITECTURES AND DATA PIPELINE

This section details the data preprocessing pipeline and the specific architectures of the deep learning models evaluated in the study. All models are implemented in PyTorch and were trained on a dataset of thermal video sequences.

A. Data Processing Pipeline

The input to the models consists of thermal video sequences. The preprocessing pipeline transforms raw video frames into structured tensors suitable for training.

1) Preprocessing (Level 0 $\rightarrow$ Level 1):

- **Input:** Raw thermal video frames (MP4).
- **Cropping:** A Region of Interest (ROI) of size 64×64 pixels is cropped around the subject center.
- **Artifact Masking:** Metal reference artifacts are identified via thresholding and masked to zero to prevent model confusion.
- **Normalization:** Pixel temperatures are normalized using statistics computed from the training set.

2) *Input Representation:* The ‘SequenceHeatmapDataset’ loader constructs the input tensor $X \in \mathbb{R}^{B \times T \times C \times H \times W}$ with:

- B : Batch Size
- T : Time steps ($T = 5$ frames)
- H, W : Spatial resolution (64×64)
- C : Input Channels (Total 5):
 - Channels 1-3: Thermal data (RGB duplicated or colormap mapped).
 - Channels 4-5: Dense Optical Flow (u, v) computed between consecutive frames to capture heat convection dynamics.

B. Model Architectures

1) *Standard Models (Scalar Regression):* These models predict a vector of 4 scalar temperature values corresponding to sensor locations.

- **SimpleResNet:** A frame-based ResNet-18 adapted for 5-channel input. Predicts 4 values per frame: $Y \in \mathbb{R}^{B \times 4}$.
- **CNN-LSTM:** Combines a ResNet encoder with an LSTM temporal aggregator. The LSTM processes the sequence of features and outputs the final temperature state: $Y \in \mathbb{R}^{B \times 4}$.

2) *Bayesian Models (Uncertainty Quantification):* To quantify predictive uncertainty, we employ Variational Inference.

- **Bayesian Neural Networks:** Standard dense layers in the regression head are replaced with ‘BayesLinear’ layers (distributions over weights).
- **Output:** Returns a tuple $(\hat{Y}, D_{KL})$, where $\hat{Y}$ is the prediction and D_{KL} is the Kullback-Leibler divergence regularization term.
- **Inference:** Uncertainty is estimated via Monte Carlo sampling (multiple forward passes).

3) Dense Prediction and Physics-Informed Models:

- **U-Net:** An encoder-decoder architecture predicting a full spatial temperature map $\hat{T}_{map} \in \mathbb{R}^{B \times 1 \times 64 \times 64}$.
- **Physics-Informed Loss (Bioheat):** Models are trained with a composite loss function:

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda_{phys} \mathcal{L}_{bioheat}$$

where $\mathcal{L}_{bioheat}$ enforces the Pennes’ Bioheat equation:

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) - \omega_b \rho_b c_b (T - T_a) + q_m + \rho c \mathbf{v} \cdot \nabla T$$

Here, the convection term $\mathbf{v} \cdot \nabla T$ utilizes the Optical Flow input channels.

